

Fabrication of Permanent Magnet by 3D Printing Technique

MinYoung Lee, Yeon Jun Choi, Ngo Tran, Jihun Ahn and BoWha Lee
Department of Physics and Oxide Research Center
Hankuk University of Foreign Studies



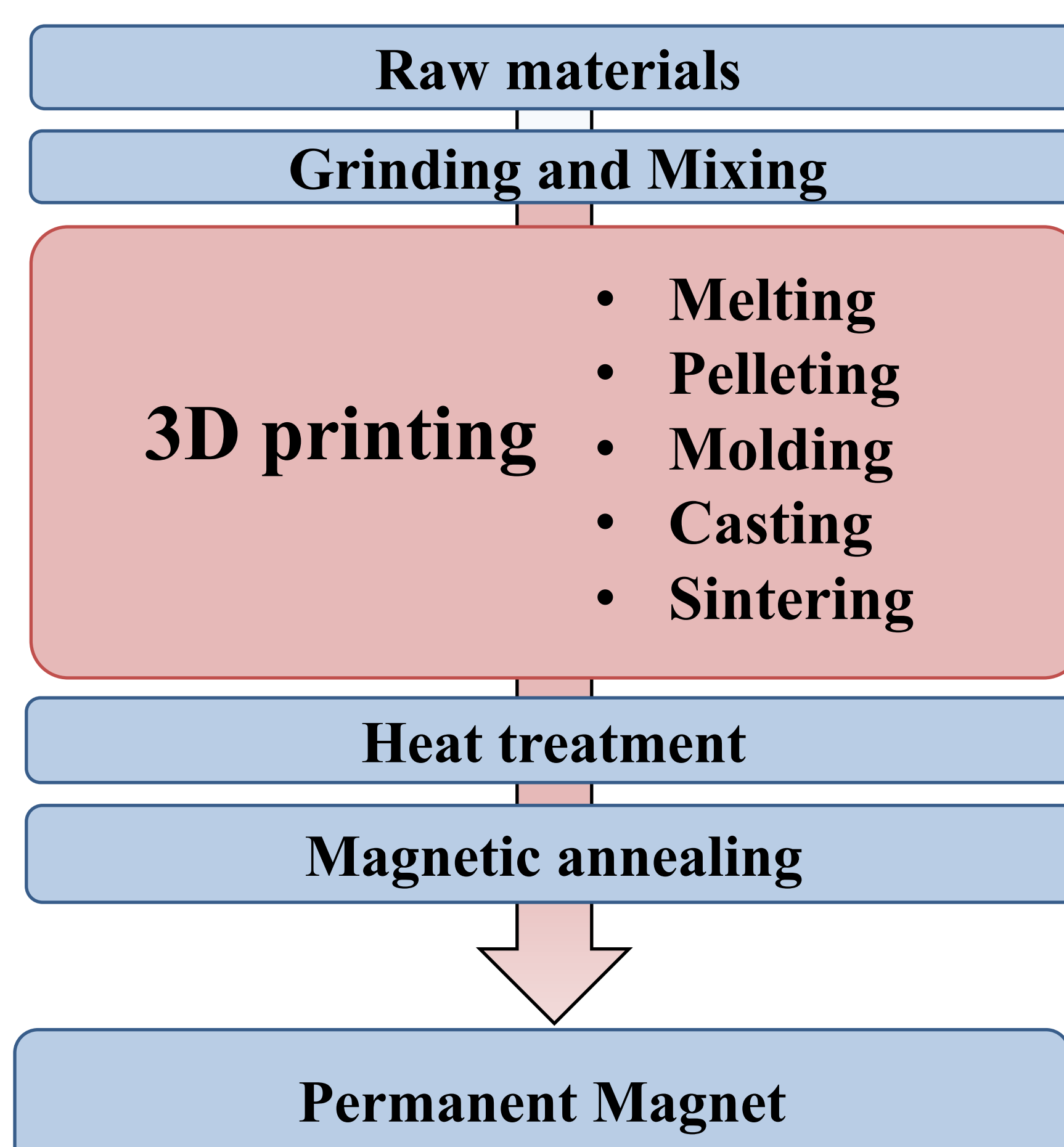
AR-09

Introduction

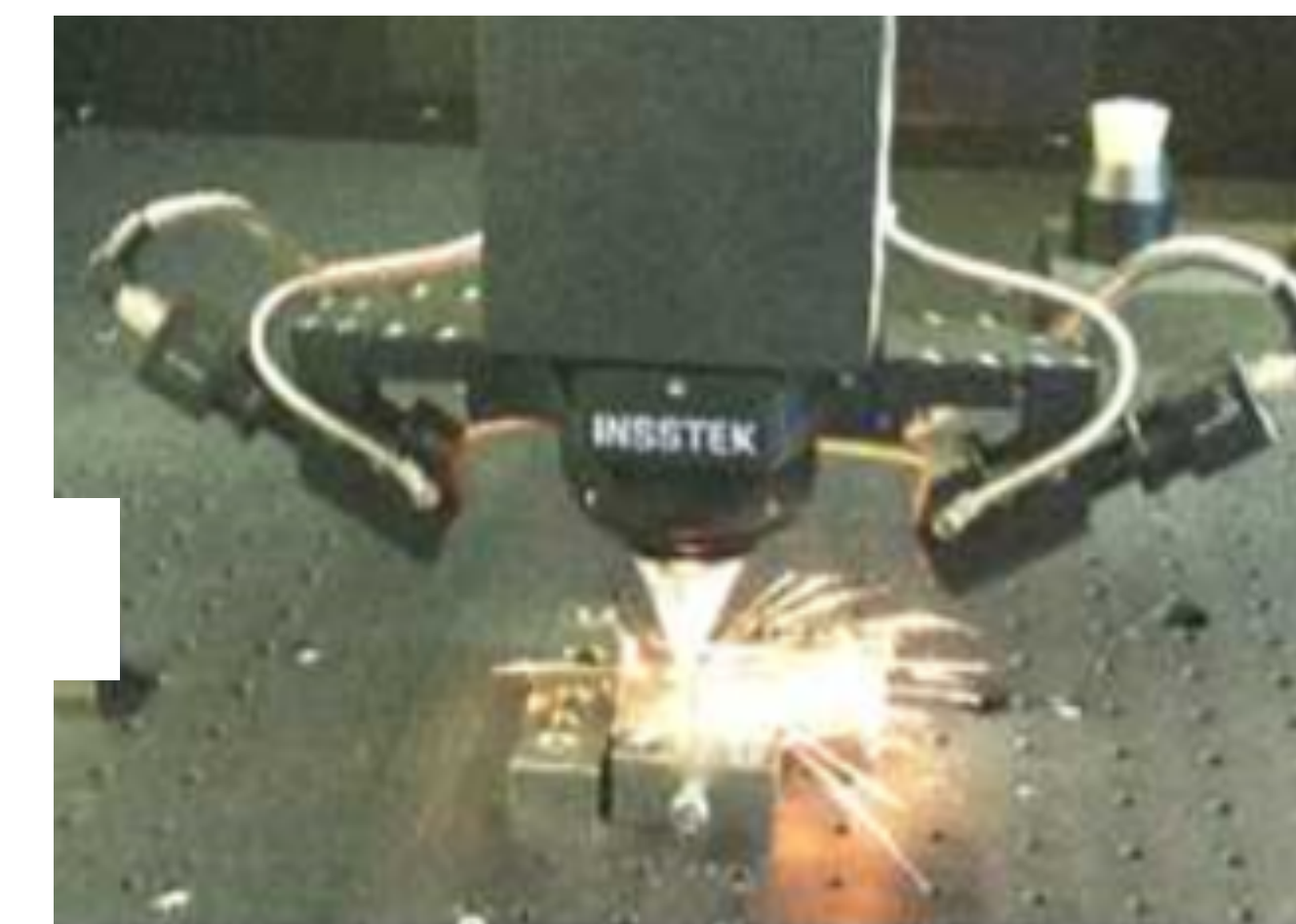
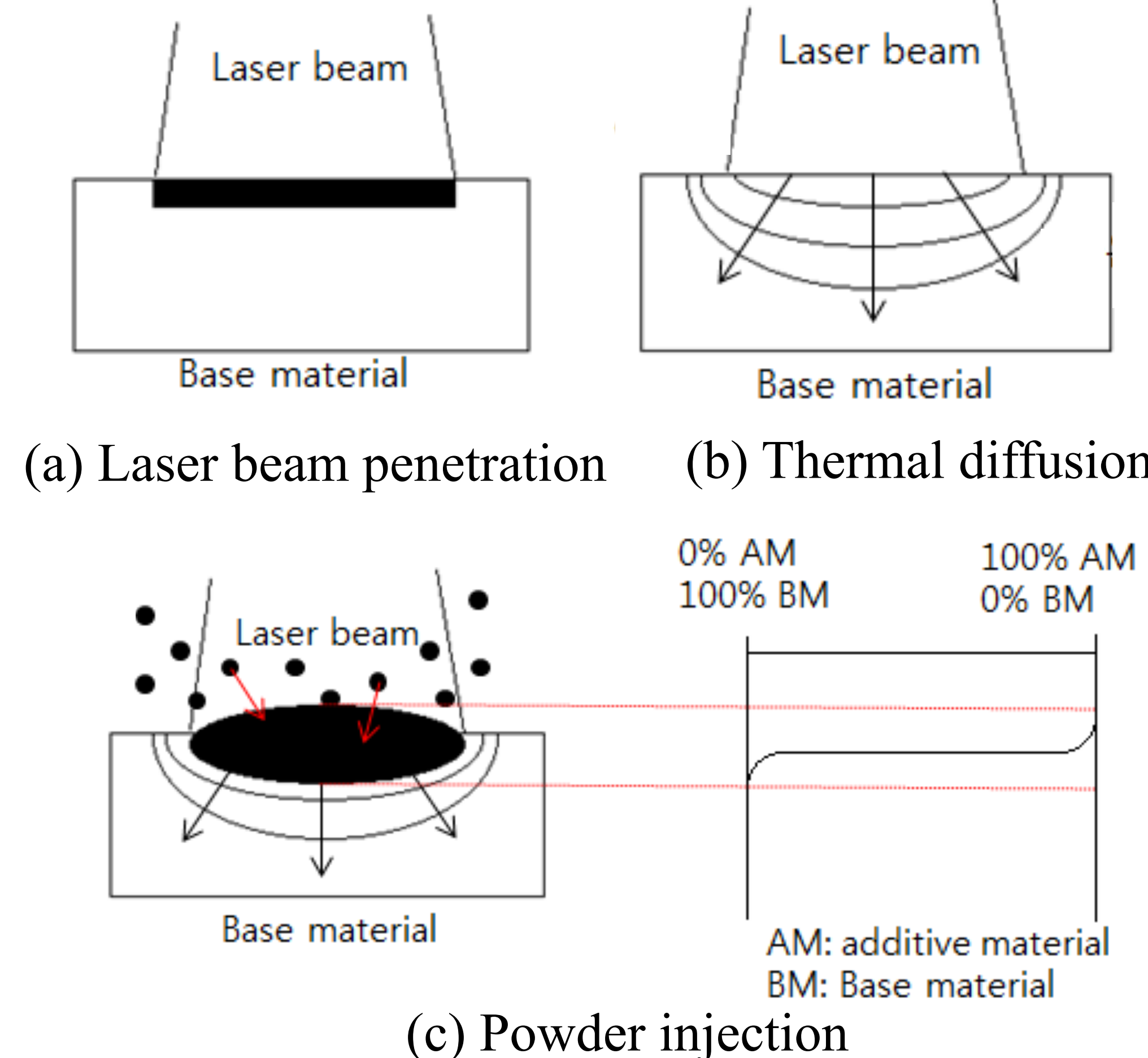
❖ Motivation

- Permanent magnets are used in a variety of industry field. These magnets are difficult to make complex shapes.
- Permanent magnet process requires large equipment such as vacuum sintering furnace, casting mold and so on.
- We can reduce the process time by using 3D printer.
- 3D printer can produce complex shapes of alloy in a short time.
- It is possible to produce flexible manufacturing cell.

❖ Process



❖ 3D printing (Laser cladding)

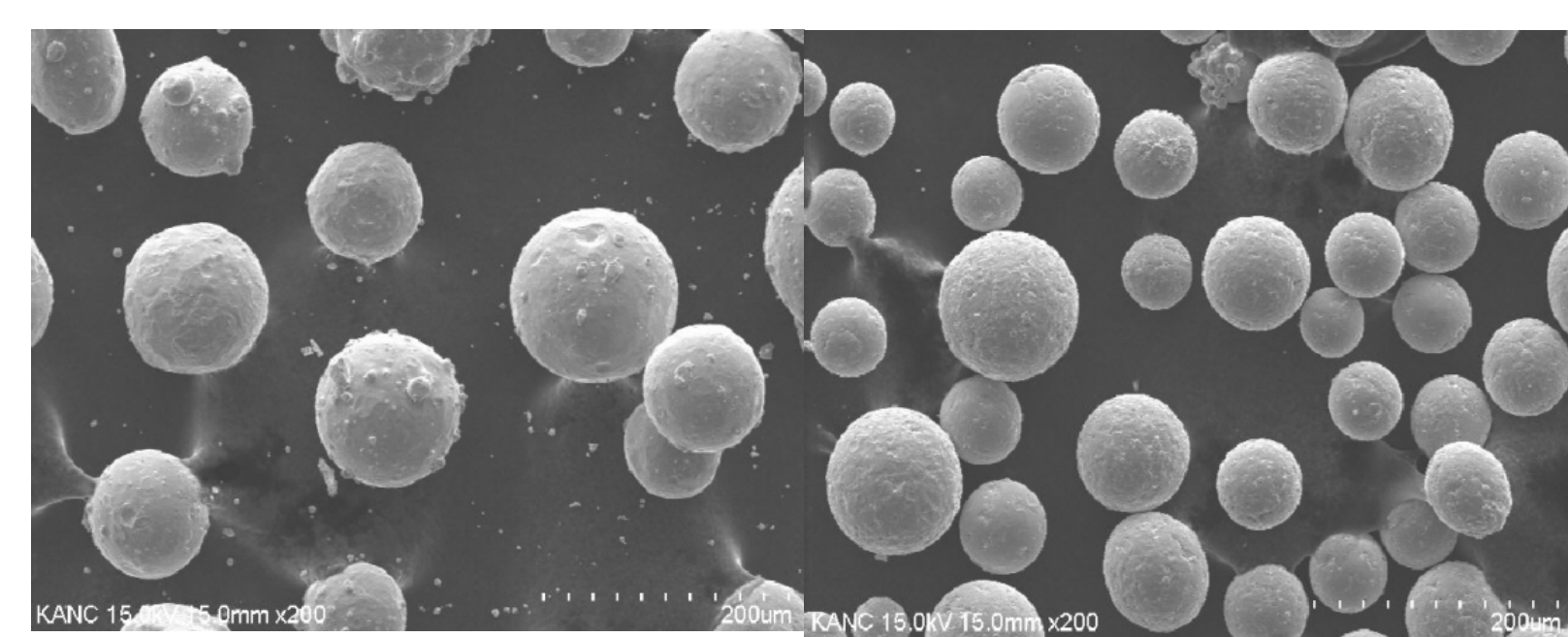


Laser cladding parameters

Laser power (kW)	2
Scanning speed (mm/min)	850
Beam diameter (mm)	1
Metal powder flow rate(g/min)	4
Metal powder particle size : 50~100 um (spherical shape)	

Experiments & Results

❖ Fabrication of cylindrical alloy rods

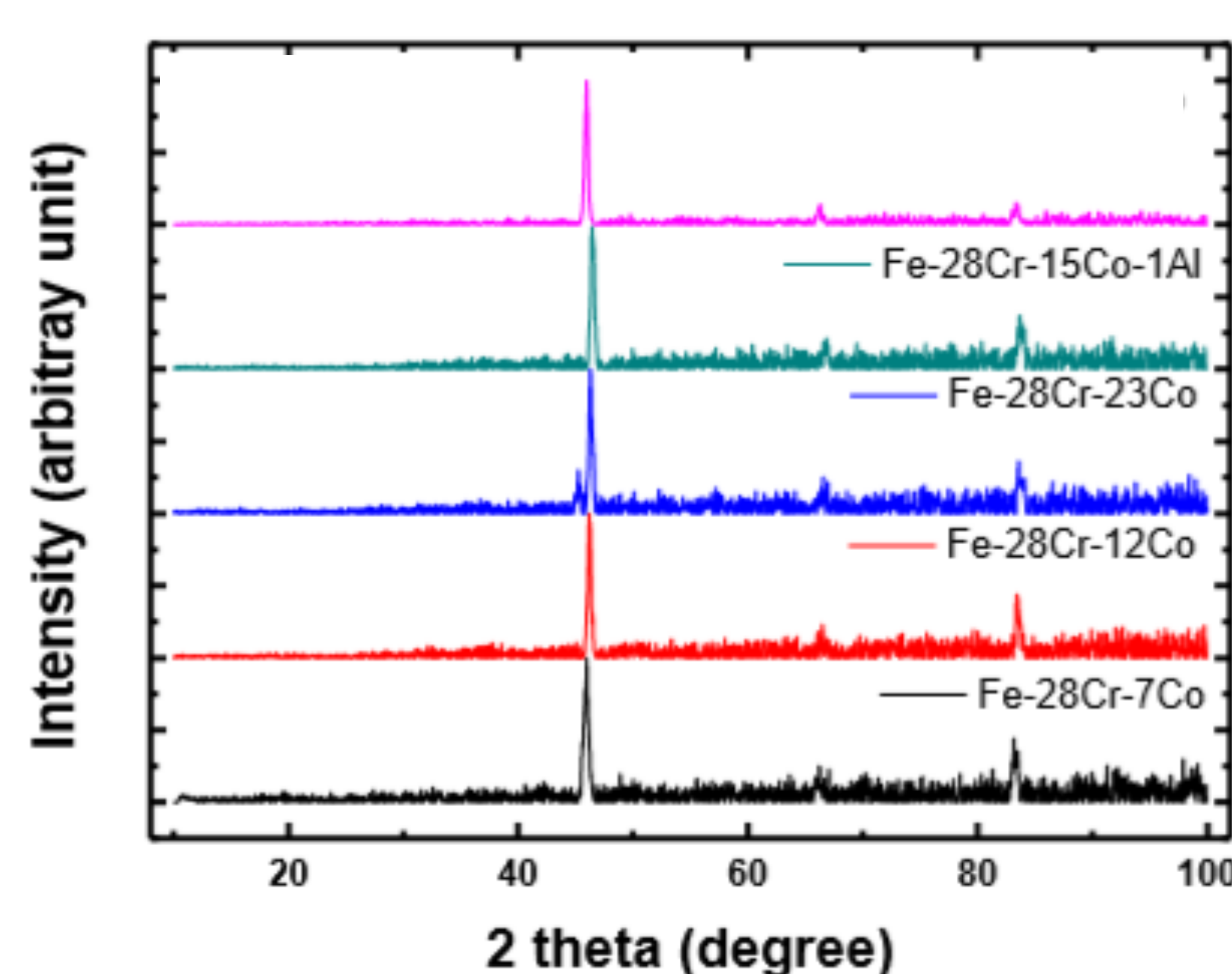


Raw materials
Spherical Iron(Fe), Chrome(Cr) powder

3D printing

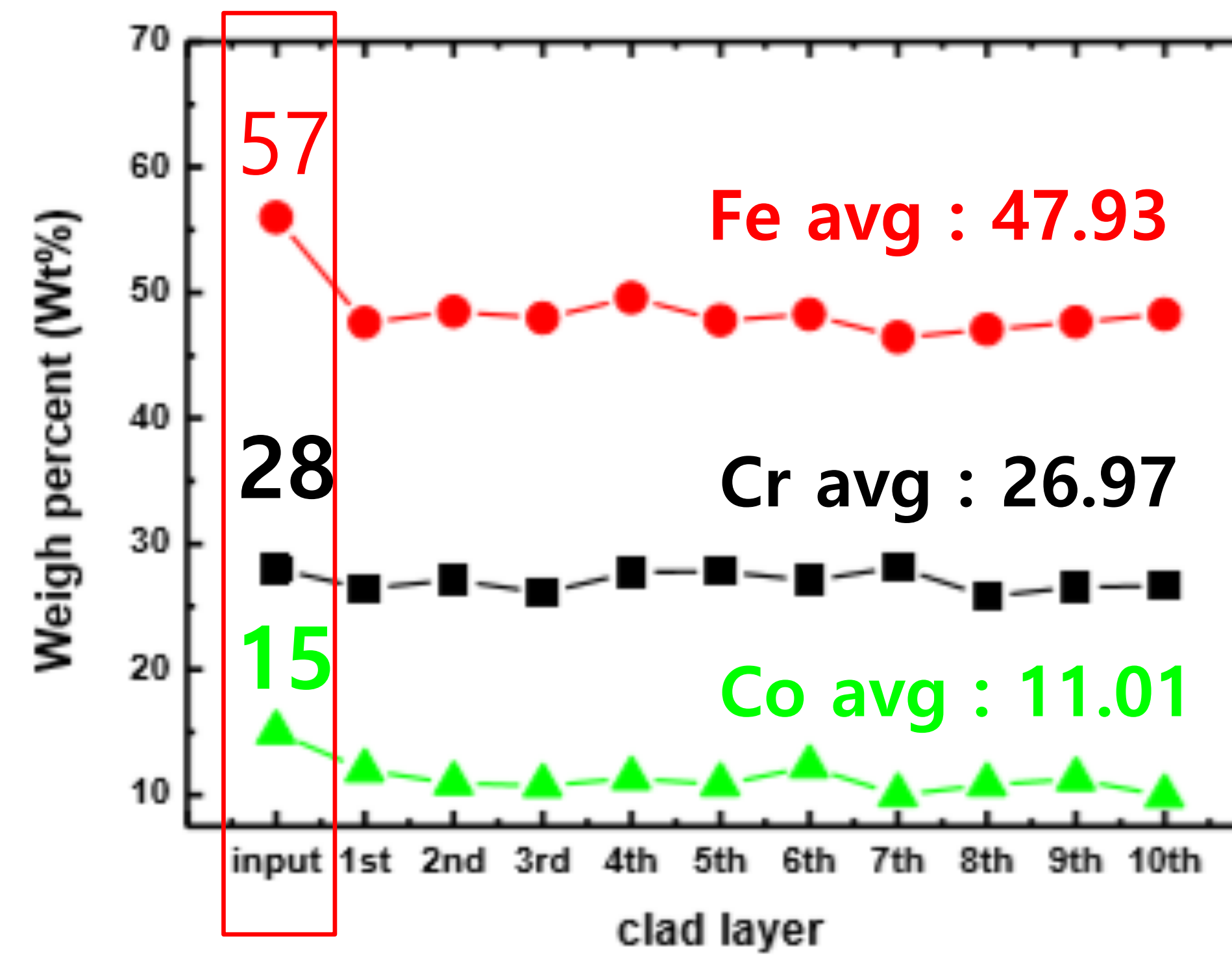


Cylindrical alloy rod
(100Fe, 57Fe-28Cr-15Co)



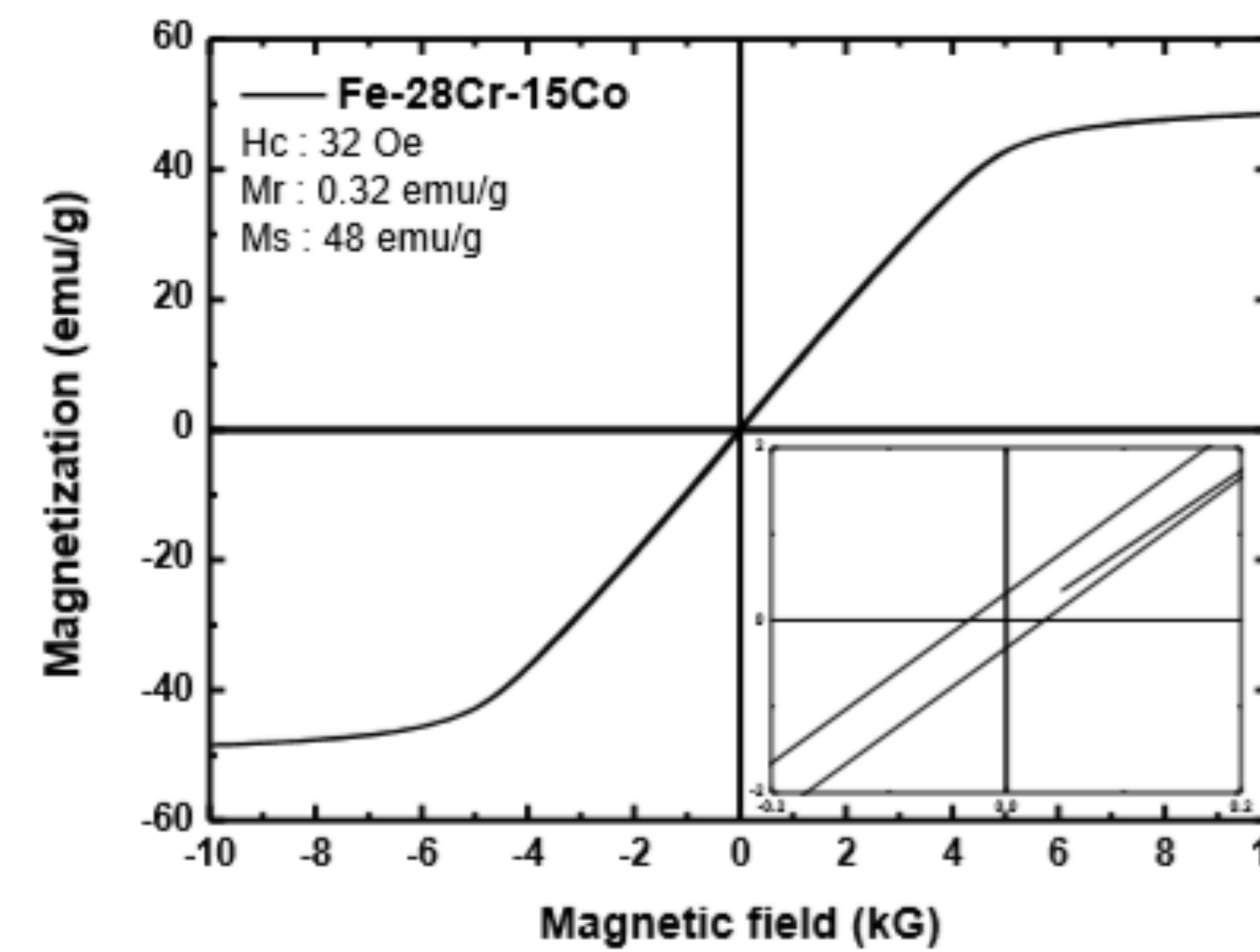
XRD : Various composite of cylindrical alloy rod

- Metal 3D printer can fabricate various composites of cylindrical alloy rod.
- XRD shows typical Fe-Cr-Co peak.



EDX : Average output composite of Fe-28Cr-15Co

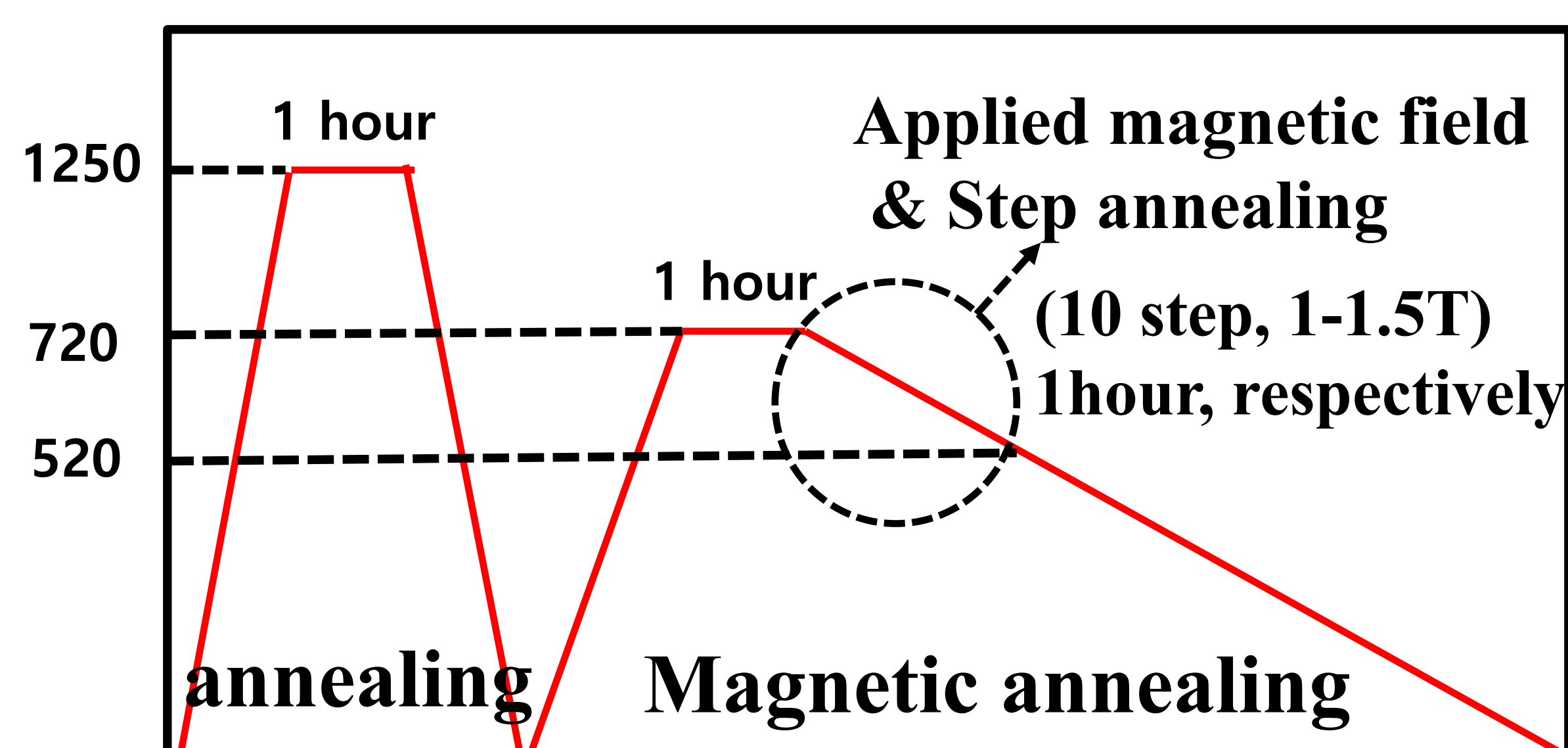
- Chemical composite of 57Fe-28Cr-15Co.
- Cylindrical alloy rods cut 10 layers to check composite of each layer.
- Powders have not same properties of density and melting point.



VSM : Hysteresis loop of initial alloy

- 57Fe-28Cr-15Co alloy shows low coercivity and low remanence.
- It can not use for permanent magnet (need for heat treatment).

❖ Heat treatment of cylindrical alloy rod



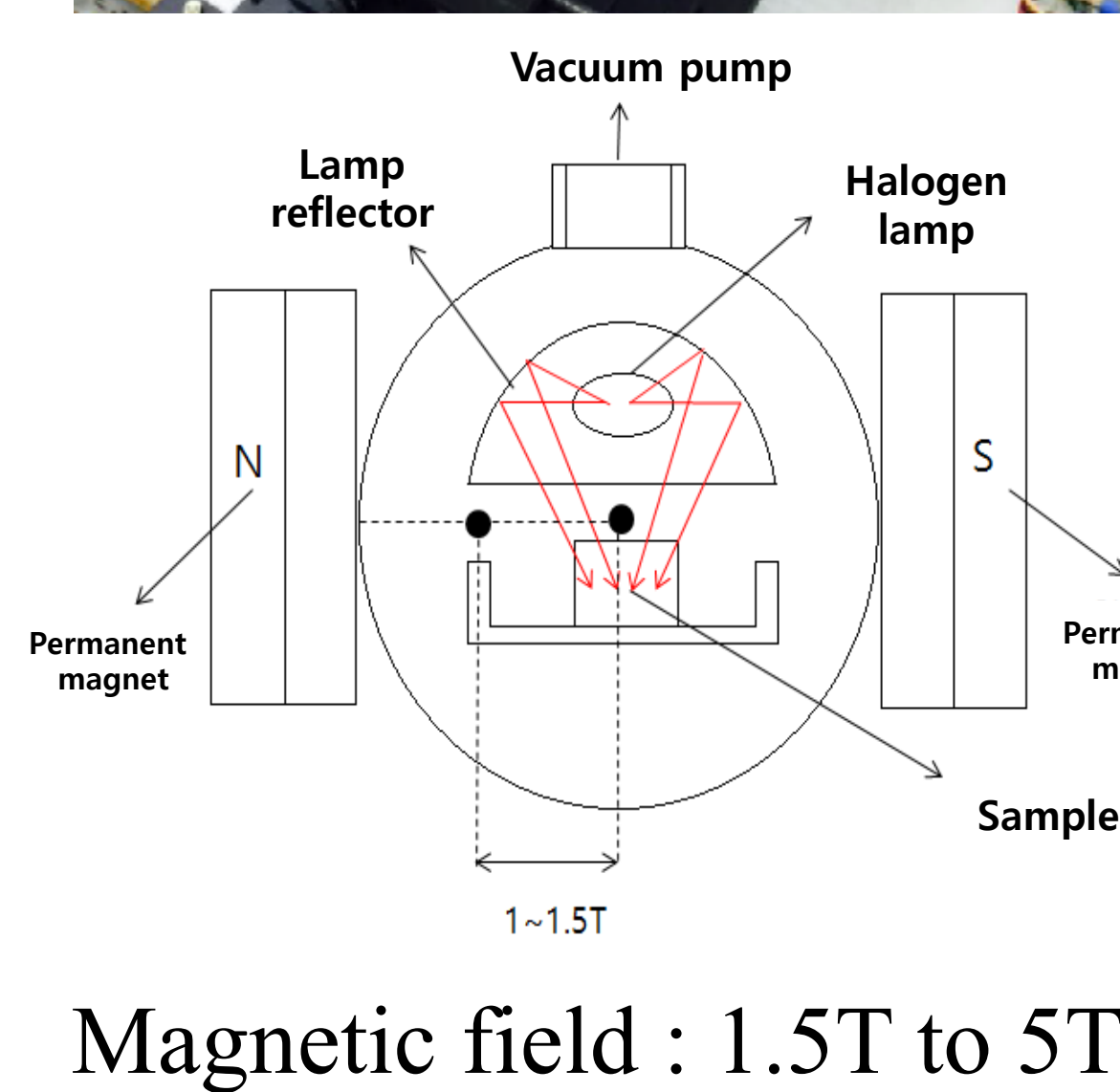
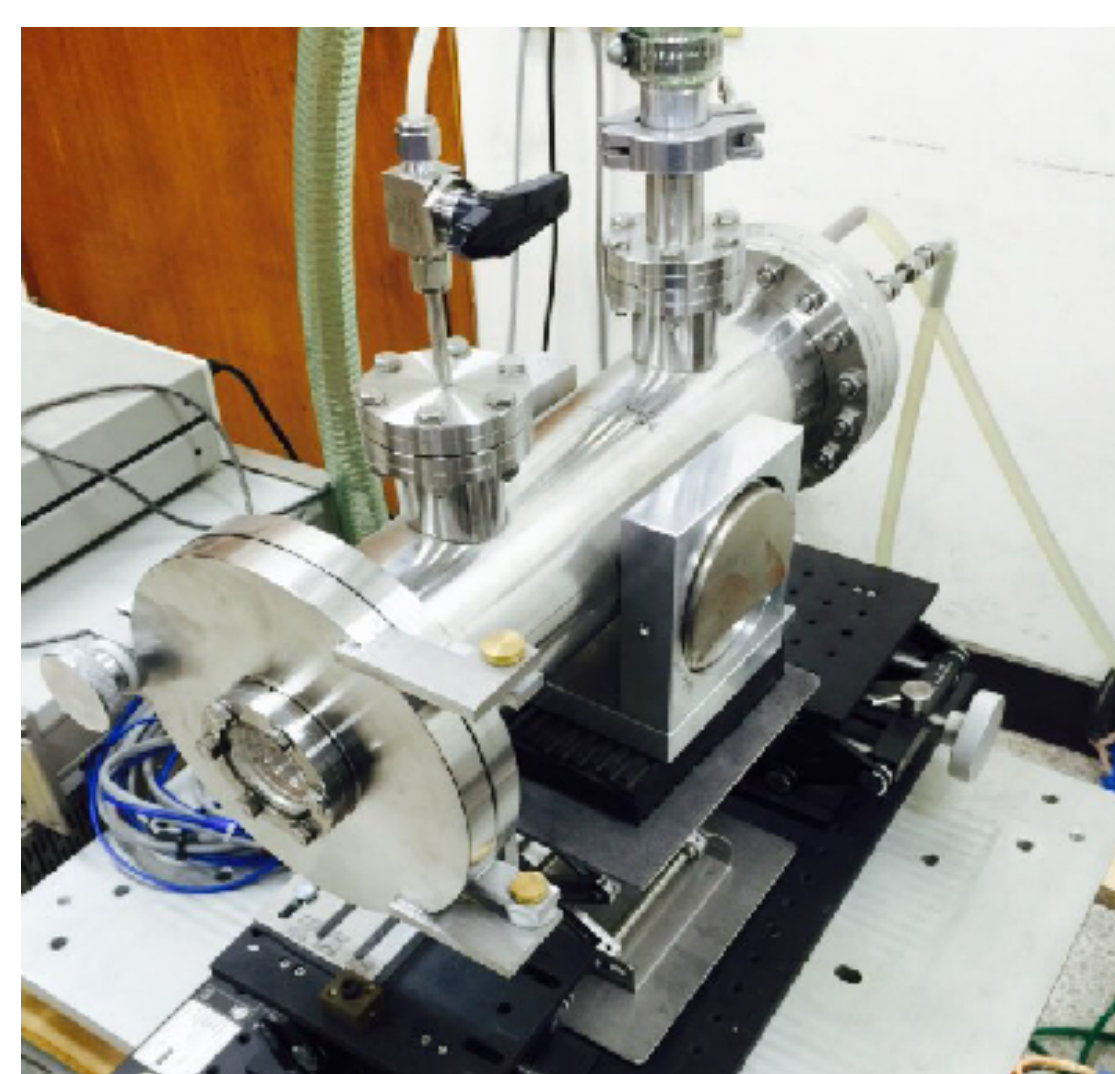
- Effect of annealing

- Uniform components by diffusion of atoms
 - Spinodal decomposition
- α_1 phase : Fe-Co rich phase (Ferromagnetic phase)
 α_2 phase : Cr rich phase

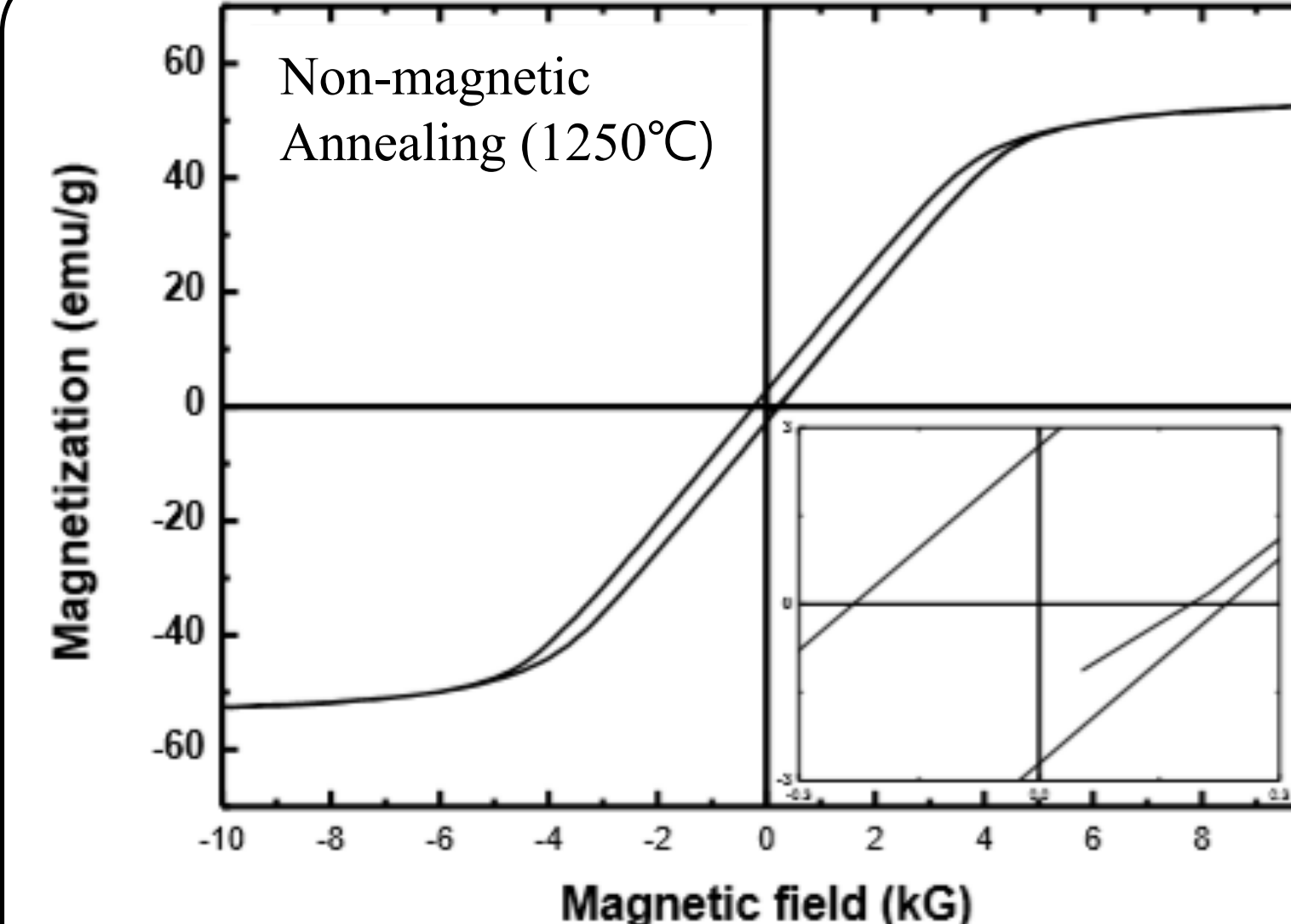
- Effect of magnetic annealing

- enhancement of magnetic anisotropy
- elongation of α_1 phase

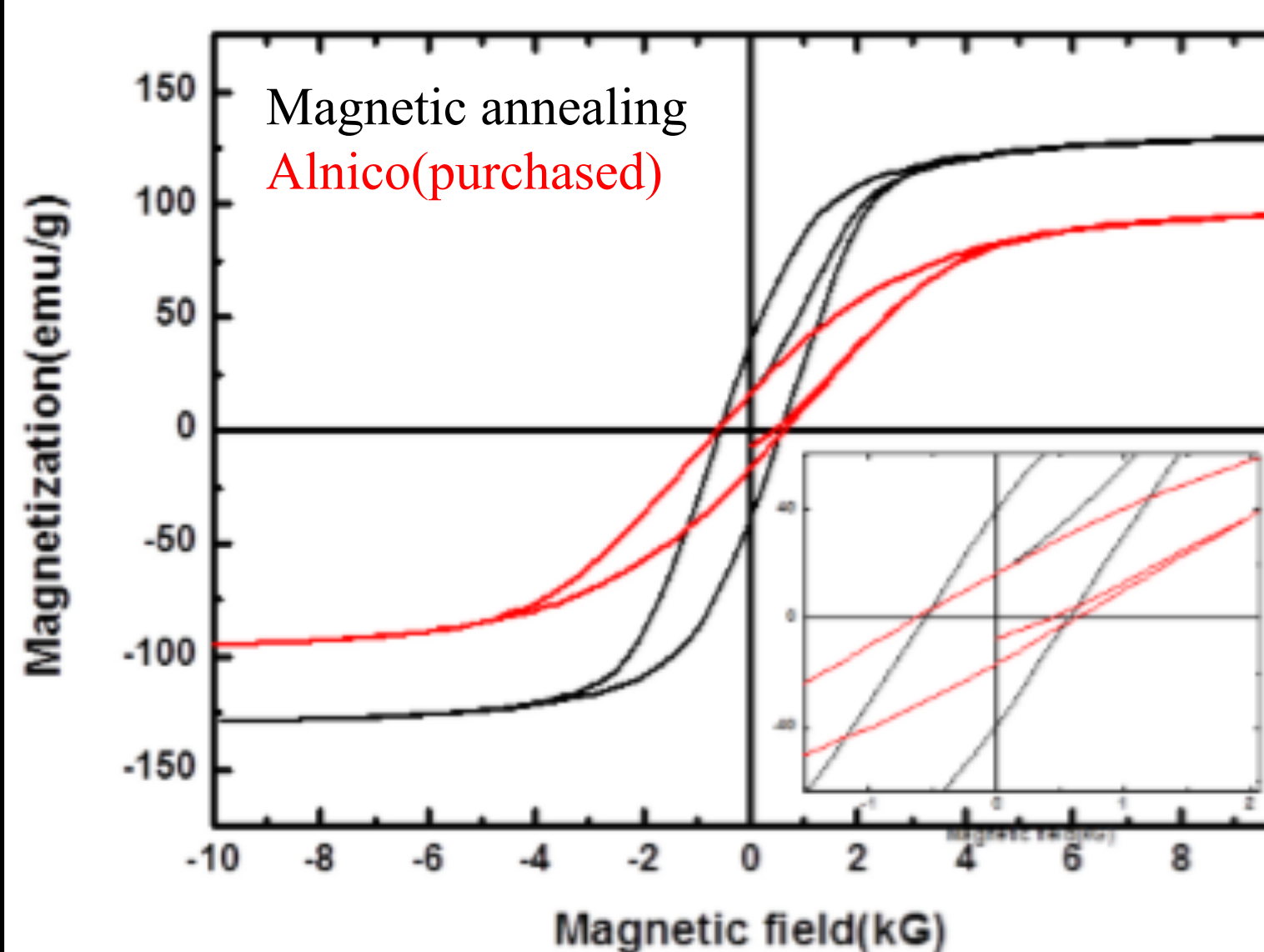
Magnetic annealing chamber



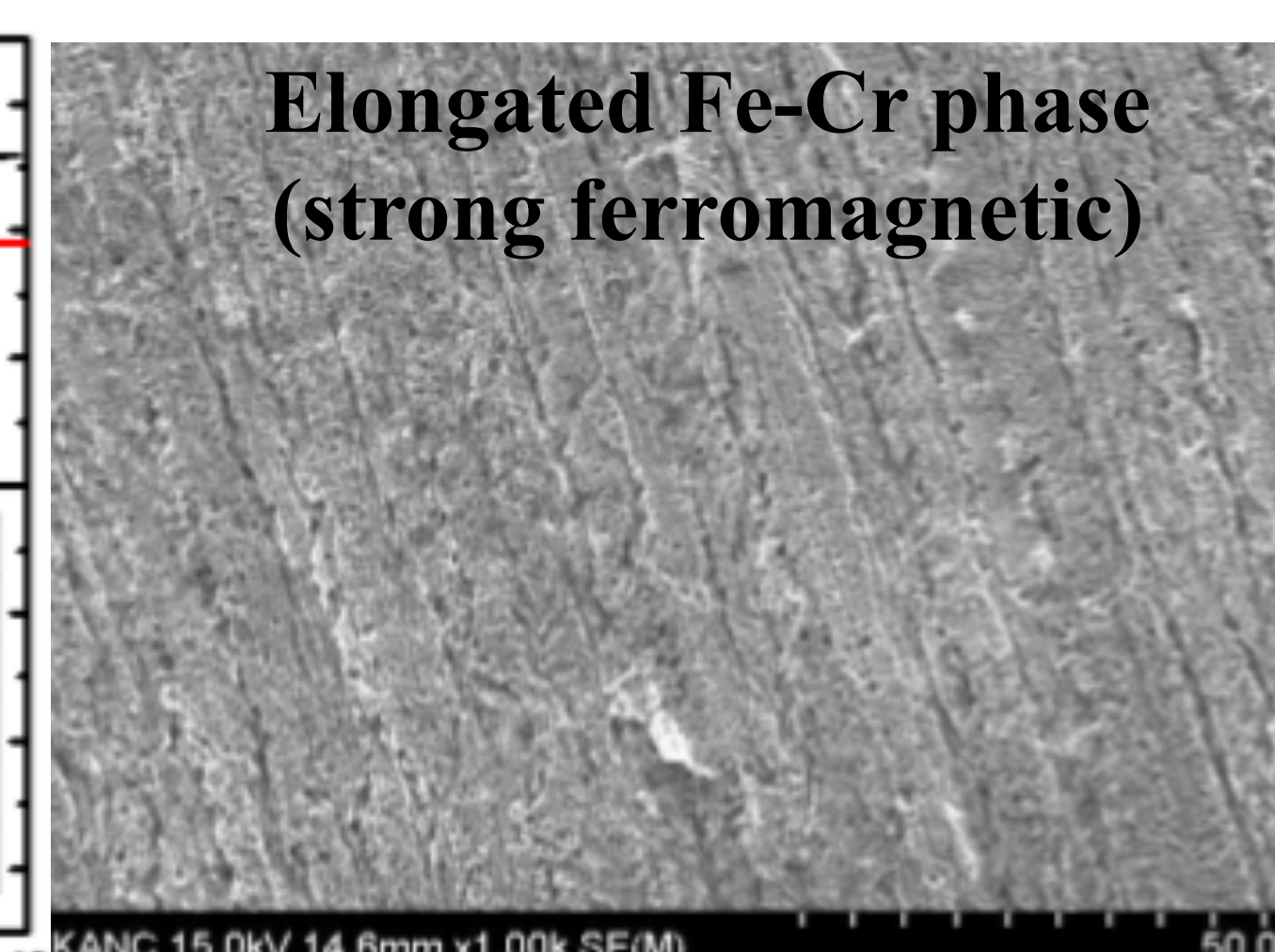
Magnetic field : 1.5T to 5T



Heat treatment (1250°C)



Magnetic annealing (10 steps)



Conclusions

- We fabricate various alloys(binary, ternary, quaternary alloy) using laser cladding technique. We can reduce the process and time for alloy fabrication.
- We confirmed the spinodal decomposition and the elongated ferromagnetic phase by heat treatment and magnetic annealing.
- We could also fabricate alloys that can be used directly in everyday life through simple surface treatment process. Therefore in terms of easy molding, the technique of laser cladding can be utilized in various related industrial applications.